Assessment 1

MIS605 Systems Analysis and Design - Requirement Analysis
Case study: Doing Good Rewards (DGR)

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Question 1: For the given case study, propose a suitable software development life cycle and project life cycle. Provide diagrams with justification.

1. Suggest one software development model and a provide diagram.

Agile methodology (specifically Scrum) is a suggested software development model. Because the DGR B2B model involves collaboration between multiple stakeholders, such as merchants, the DGR team, and customers, it also requires adaptability to evolving requirements.

Agile prioritises rapid iterations, continuous feedback, and user-centric design, making it suitable for such dynamic environments.

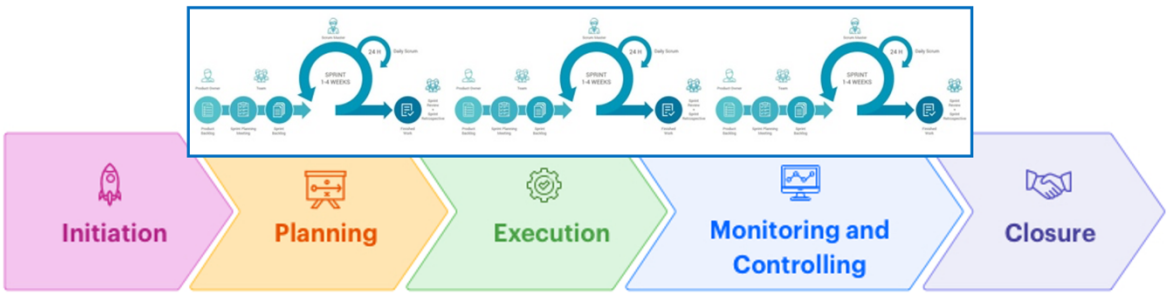
The use case diagram helps visualise the interactions between the DGR platform and various actors during invoice processing, discount calculation, and donation facilitation.

2. Discuss Project Life Cycle

The DGR B2B model project follows a structured Project Life Cycle with these key phases:

1. **Initiation:** Define scope, feasibility, stakeholders, team, and a high-level plan.
2. **Planning:** Gather detailed requirements, design system architecture, create WBS, and refine the project plan with risks considered.
3. **Execution:** Develop the platform using Agile methods, conduct testing, manage resources, and address changes.
4. **Monitoring and Control:** Track progress, identify issues, report to stakeholders, and have project reviews.
5. **Closing:** Deliver the platform, evaluate success, document lessons learned, and formally close the project.

3. Mapping of your software development methodology on project life cycle



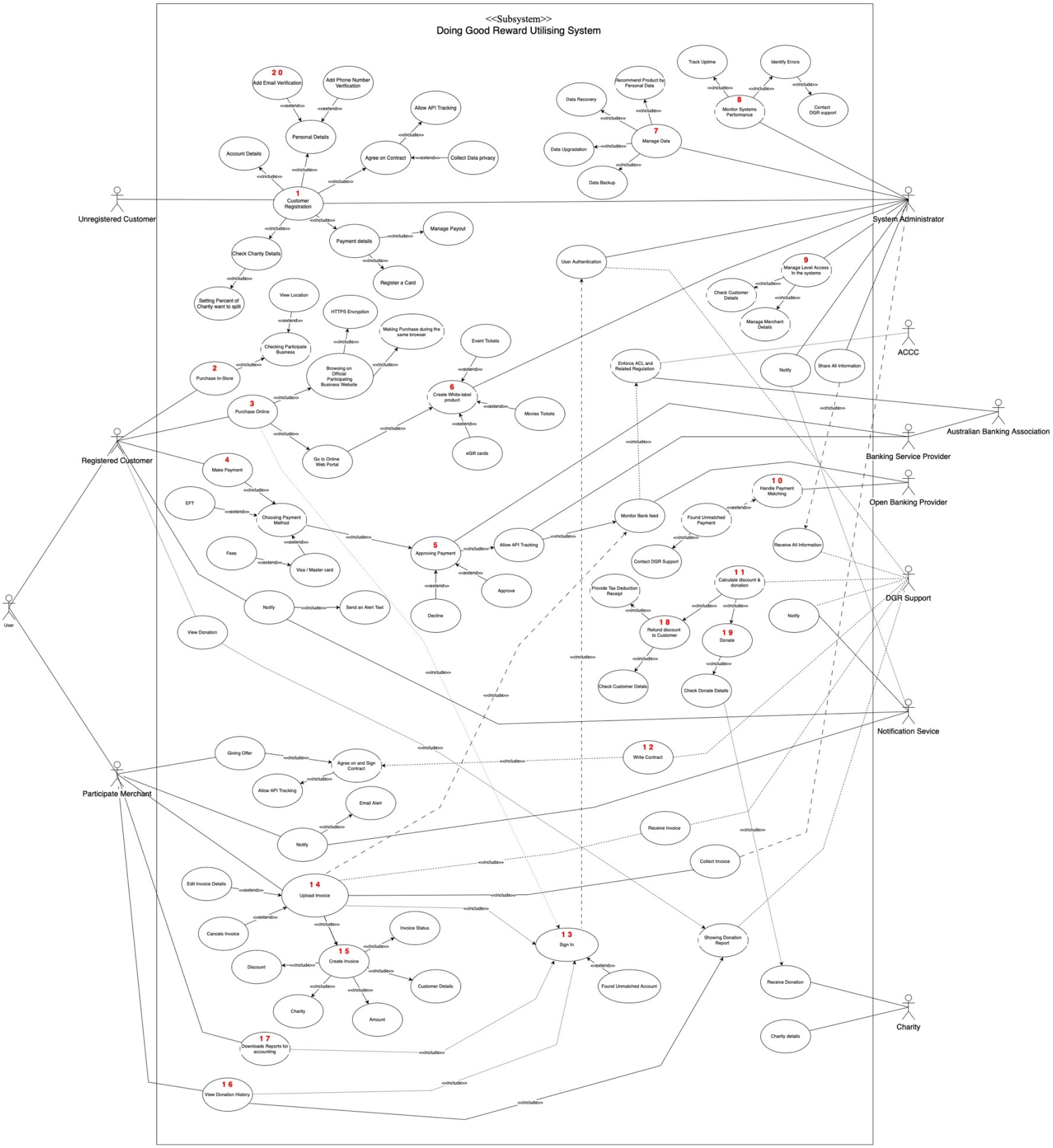
1. **Initiation:**
   - Define project scope, goals, and feasibility.
   - Secure project funding and establish a project team with developers, testers, DGR admins, and a project manager.
   - Create a high-level project plan.
2. **Planning & Design:**
   - Conduct in-depth requirement gathering and analysis.
   - Design system architecture, user interface (UI) mockups, and data flow diagrams.
   - Define user roles, access levels, and functionalities in detail.
   - Develop a detailed project plan with sprints, user stories, and assigned tasks.
3. **Development & Testing:**
   - Implement functionalities in agile sprints (e.g., two-week iterations) using Scrum ceremonies.
   - Prioritise core functionalities for initial releases, add others in subsequent sprints.
   - Employ Open Banking APIs for secure bank data access with Merchant consent.
   - Conduct unit testing, integration testing, security testing, and user acceptance testing with Merchant.
4. **Deployment & Launch:**
   - Release the DGR platform in phases:
     - **Initial version:** Core functionalities (invoice processing, discount calculation)
     - **Subsequent releases:** Additional features
   - Train Merchant on system usage and functionalities.
   - Initiate marketing and outreach efforts to attract businesses.
5. **Monitoring & Maintenance:**
   - Monitor system performance, user activity, and error logs.
   - Address bugs, security patches, and feature enhancements based on feedback and usage data.
   - Stay updated on Open Banking regulations and adapt the platform accordingly.
6. **Closure:**
   - Evaluate project success against objectives, gather lessons learned, and conduct a post-mortem analysis.
   - Document the project and transition ownership to the DGR operations team.

Question 2: Identify all the human and the non-human actors within the system. Provide their type and brief description against every actor.

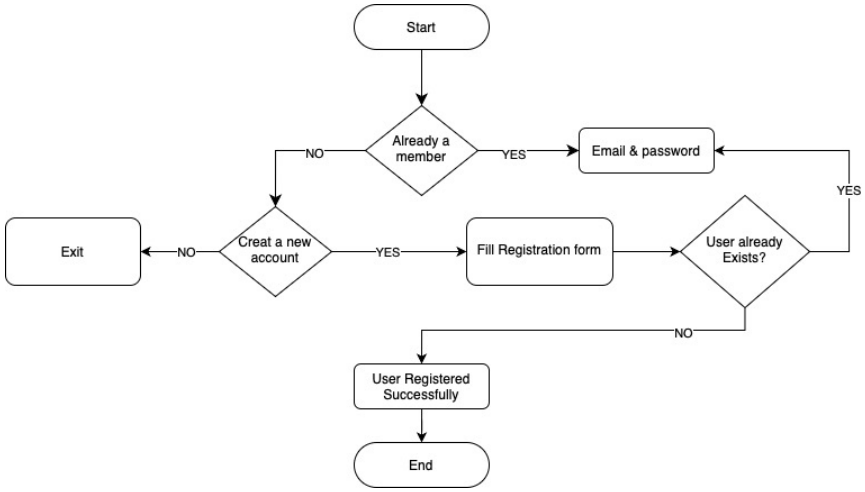
Actor Name	Type of the actor	Description
Participating Merchant	Primary	The main user who interacts with the DGR platform to manage discounts and donations, track programme activity, upload invoices, and interact with registered customers.
Registered Customer	Primary	They interact with the DGR system by creating an account under a registered contract and making payments to merchants for invoices processed through the DGR platform.
Unregistered Customer	Secondary	Customers who are not yet registered who are interested and stay in the registration process or who may pass into the system to see the benefits and value of this activity, might be an incentive to register in the future.
DGR Support	Secondary	Solving the problem by providing technical or account-related assistance to the users who encounter issues with the DGR platform.
System Administrator	Secondary	Oversees overall platform operations, manages user accounts, system settings, and collects data
Open Banking Provider	Secondary	A third-party service that facilitates secure access to participating merchants' bank and customers' bank account data for transaction verification purposes.

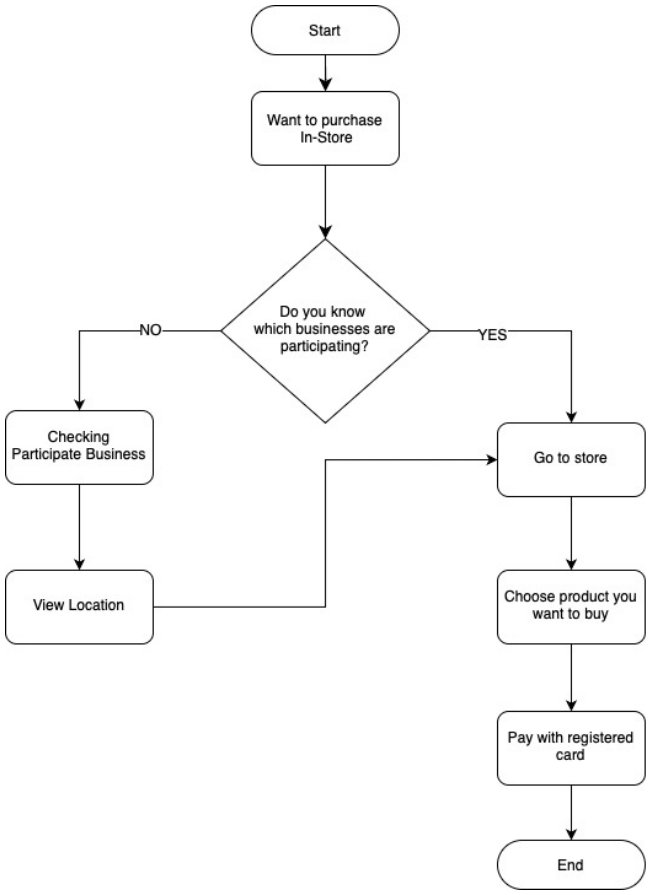
Banking Service Provider	Secondary	Companies that handle electronic payments, such as credit card transactions and EFT. Also, it gives permission and facilitates, after the customer's consent to enable third parties to securely access customer financial information
Charity	Secondary	The organisation that receives donations facilitated by the DGR platform based on calculated amounts from processed invoices.
Email Delivery Service	Secondary	An external service that transmits notifications from the DGR platform to various actors, for instance, alert text to customers, email to merchants.
DGR Platform	Abstract	The software application that manages the DGR B2B model functionalities, including invoice processing, discount calculations, data storage, and interactions with other actors.
Data Storage	Abstract	The system that securely stores data related to invoices, transactions, discounts, donations, user accounts.

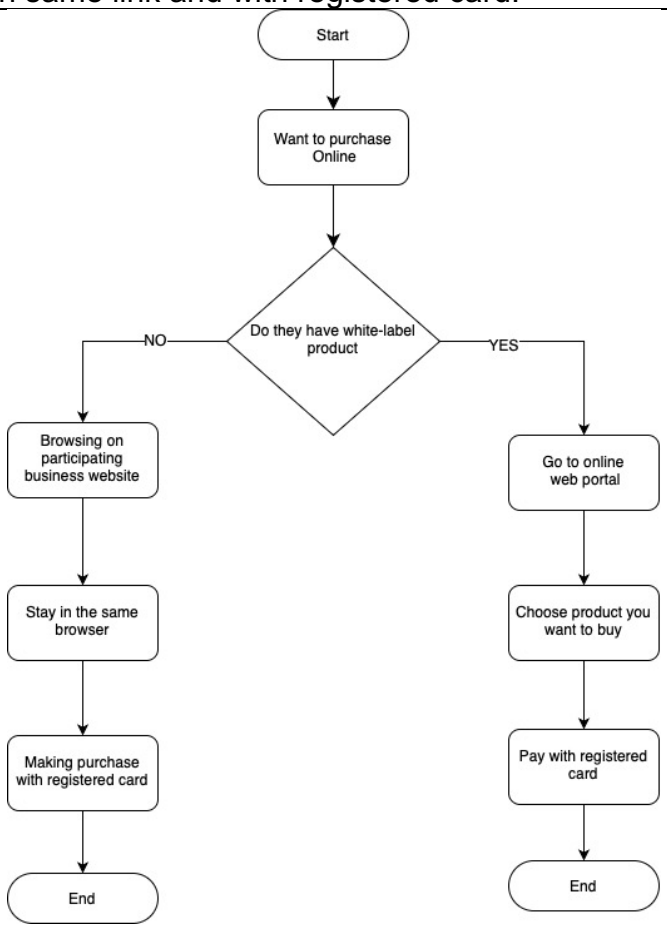
Question 3: Use Case Diagram

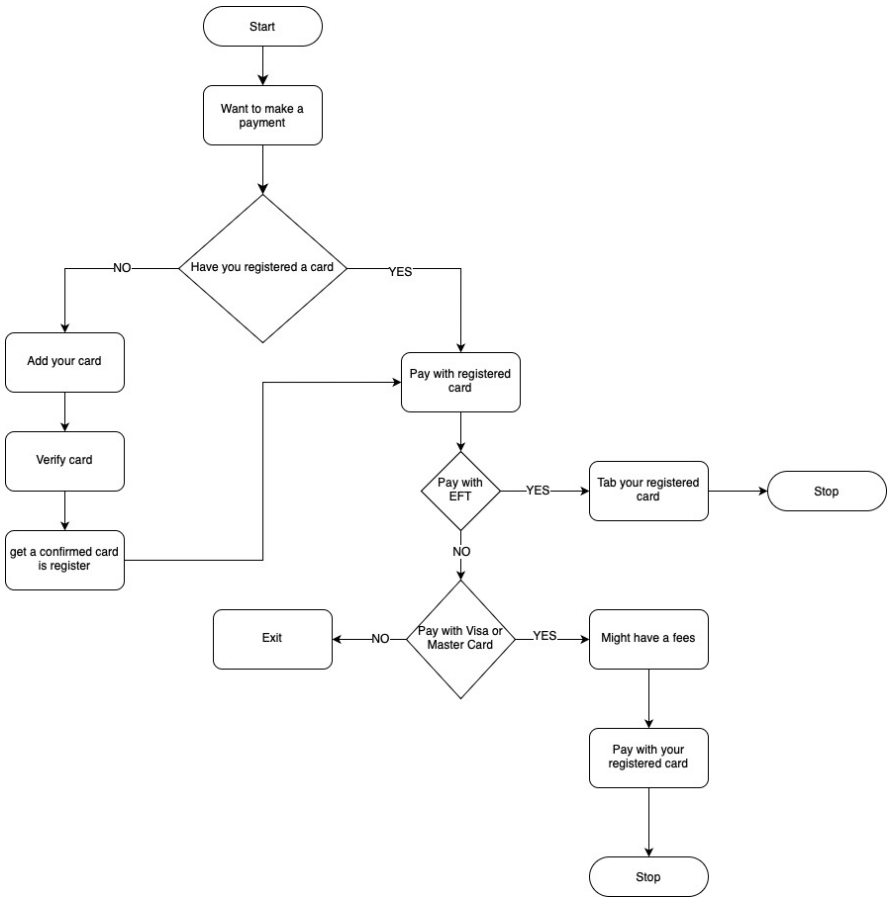


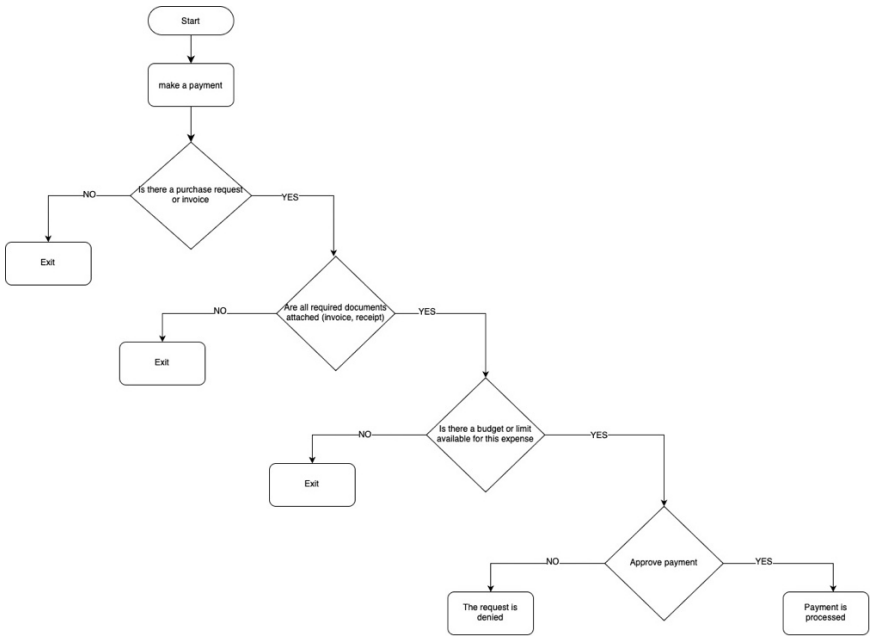
Question 4

Use Case Number	1
Use Case Type	Base
Use Case Name	Customer Registration
Actor Involved	Unregistered customer, System Administrator and DGR Platform
Associated Data Sources	DGR DB
Associated User Interfaces	Customer Registration
Pre-Condition(s)	User account subsystems must be available
Post Condition(s)	Customers' account and details must be saved in the system once the information is authenticated by system administrator
Detailed Description	Customer desire to create an account and provides the basic personal details, account details, chosen charity, agree on contact, and payment details to accept offer from business. After verification of the credit/debit card details, system creates an account for the customer as registered customer.
Normal Course of Events Using a Flow Chart Diagram	
Alternate Course(s)	If the customer already has an account by using invalid email, the system informs them, a new registration cannot be created please check your details.

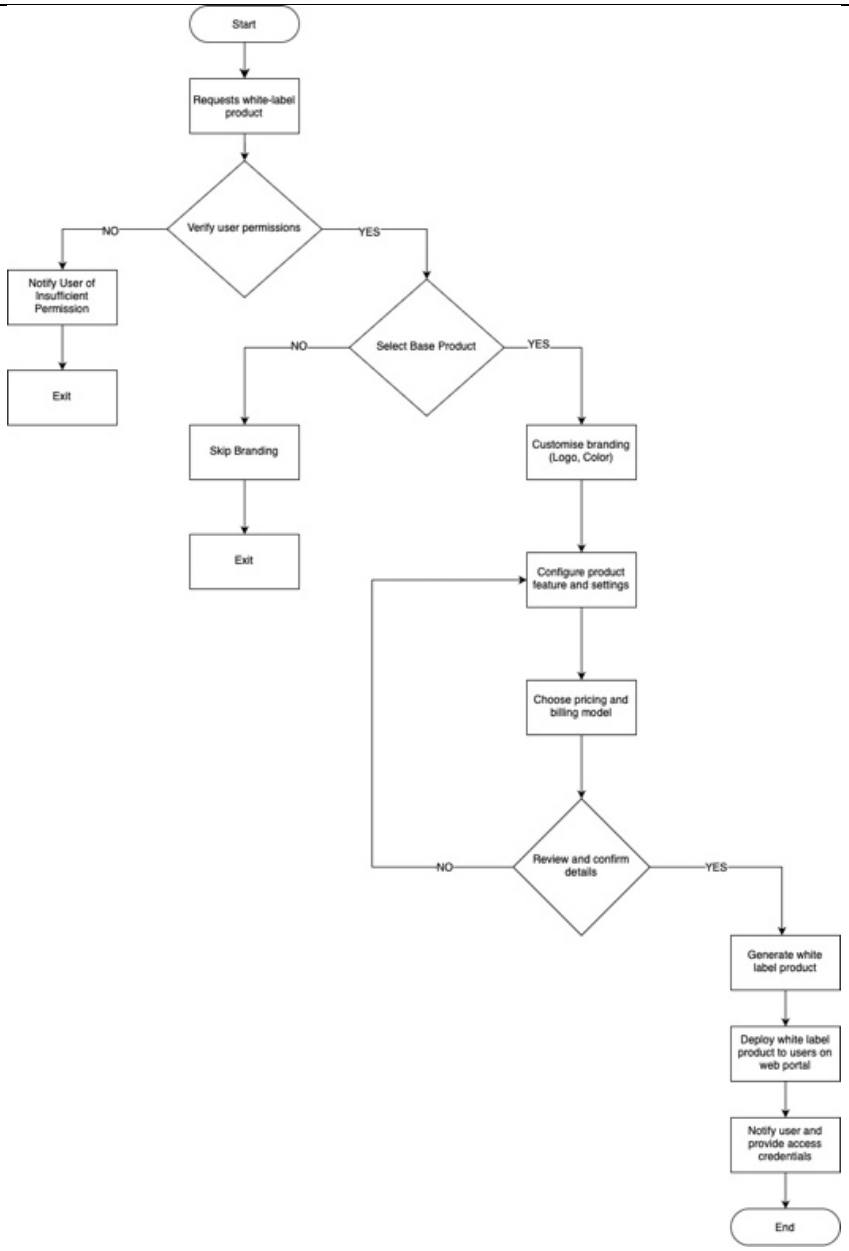
Use Case Number	2
Use Case Type	Base
Use Case Name	Purchase In-Store
Actor Involved	Registered customer, DGR Platform and Merchant
Associated Data Sources	DGR DB
Associated User Interfaces	Purchase In-Store
Pre-Condition(s)	Customer is a registered user on the DGR platform, and has a valid payment method linked to their account
Post Condition(s)	Invoice is uploaded by the participating merchant to the DGR platform, then discount will be calculated based on customer's profile to set how much they want to donate, and system records the transaction and associated discount in the database
Detailed Description	A registered customer wants to make a purchase at the store. If they didn't know which business is participating, they could check on the DGR application, also map location. Then, go to store and purchase with registered card, which fees might apply.
Normal Course of Events Using a Flow Chart Diagram	
Alternate Course(s)	If the customer hasn't paid with registered card, they won't get a refund discount and donate to charity from the merchant offer.

Use Case Number	3
Use Case Type	Base
Use Case Name	Purchase Online
Actor Involved	Registered customer, DGR Platform and Merchant
Associated Data Source	DGR DB
Associated User Interfaces	Purchase Online
Pre-Condition(s)	Customer is a registered user on the DGR platform and has a registered payment linked to their account. Also, Participating merchant's account is active on the DGR platform
Post Condition(s)	Full payment for the invoice is received by the merchant. After payment matching process, discount amount will be calculated based on customer profile. If a discount is applicable, the DGR platform will notify by text and initiate a refund to the customer account. Also, all details will be stored in the customer's account and merchant's account on DB.
Detailed Description	The customer selects a participating merchant if the products show with white label, they must go to purchase on online web portal. If not, just browse their main website, and must pay within same link and with registered card.
Normal Course of Events Using a Flow Chart Diagram	
Alternate Course(s)	If the customer hasn't paid with registered card, they won't get a refund discount and donate to charity from the merchant offer.

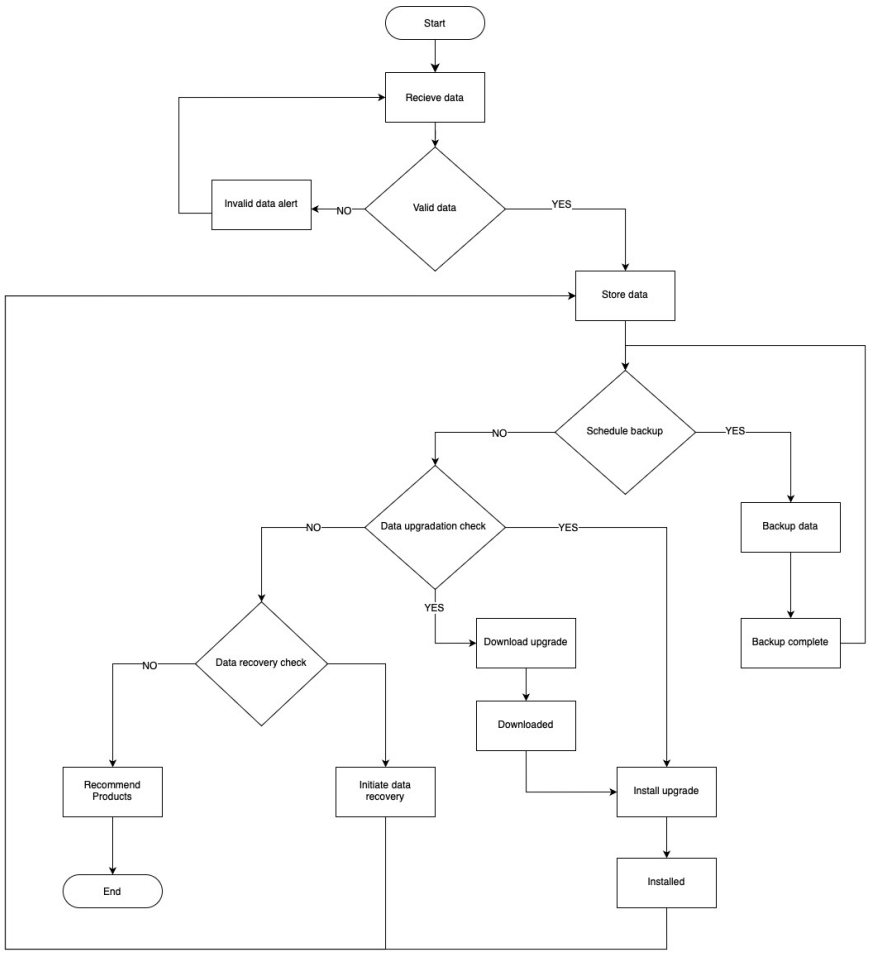
Use Case Number	4
Use Case Type	Base
Use Case Name	Make Payment
Actor Involved	Registered customer, DGR Platform and Merchant
Associated Data Source	DGR DB
Associated User Interfaces	Make Payment
Pre-Condition(s)	Customer is a registered user on the DGR platform and has a registered payment linked to their account.
Post Condition(s)	Full payment for the invoice is received by the merchant. Discount amount will be calculated and divide to return to the customer and chosen charity. Invoice status is updated to “Paid” in the system, and customers will receive confirmation of successful payment.
Detailed Description	Credits the full invoice amount to the merchant by registered card. If you don’t have registered card, you must verify your card before making a payment. Fees might apply to Visa and Mastercard payment methods.
Normal Course of Events Using a Flow Chart Diagram	
Alternate Course(s)	If the payment authorization fails due to insufficient funds or invalid card information, the system informs the customer and allows them to retry with a different payment method.

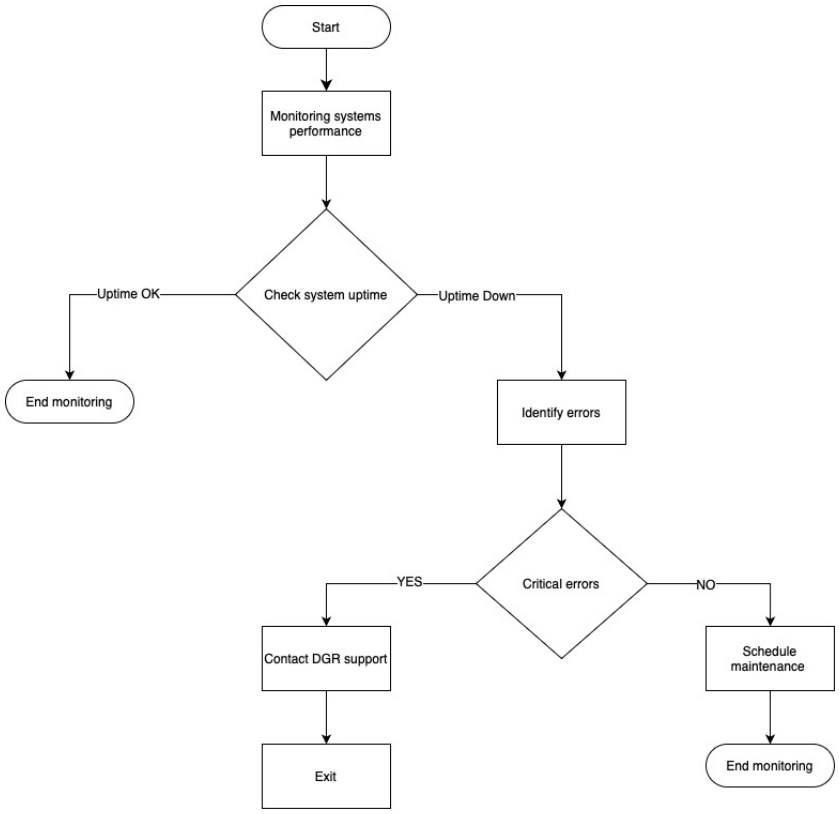
Use Case Number	5
Use Case Type	Include
Use Case Name	Approving Payment
Actor Involved	Registered customer, DGR Platform, Merchant, and Bank Service
Associated Data Source	DGR DB
Associated User Interfaces	Approving Payment
Pre-Condition(s)	Customer is a registered user on the DGR platform and has a registered payment linked to their account.
Post Condition(s)	Full payment for the invoice is received by the merchant. Discount amount will be calculated and divided to return to customer and chosen charity. Invoice status is updated to “Paid” in the system, and customer will receive confirmation of successful payment.
Detailed Description	Verifies invoice details and customer information, if the payment method used is within the limit available and sufficient funds, it will be approved by bank service.
Normal Course of Events Using a Flow Chart Diagram	
Alternate Course(s)	If the payment authorisation fails due to insufficient funds or invalid card information, the system shows denied and informs the customer and allows them to retry with a different payment method.

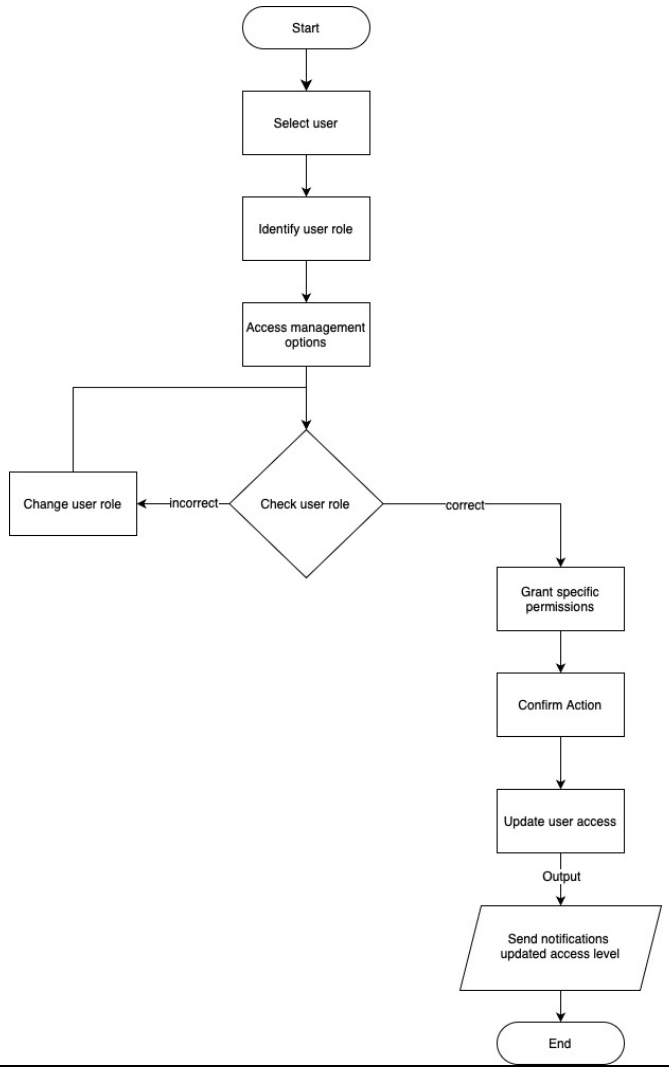
Use Case Number	6
Use Case Type	Include
Use Case Name	Create White-Label
Actor Involved	Registered customer, DGR Platform, and Merchant
Associated Data Source	DGR DB
Associated User Interfaces	Create White-Label
Pre-Condition(s)	System administrator account must be active and accessible. And DGR platform infrastructure must be operational.
Post Condition(s)	A new white-labelled instance of the DGR platform is created for the participating business. Business branding and configuration details are successfully applied to the white-labelled instance. The business receives login credentials to access their white-labelled DGR platform.
Detailed Description	A new business desires to participate in the DGR programme and offer discounts to their customers. They provide their business information, branding assets (logos, colour schemes), and any desired configuration settings to the DGR platform administrator. The DGR platform administrator configures a new white-labelled instance of the DGR platform with the provided business branding and settings. Once configuration is complete and verified, the administrator creates login credentials for the business and grants them access to their white-labelled platform.
Normal Course of Events Using a Flow Chart Diagram	

	
Alternate Course(s)	If the business already has a white-labelled instance associated with them, the system informs the administrator and suggests contacting support for further assistance.

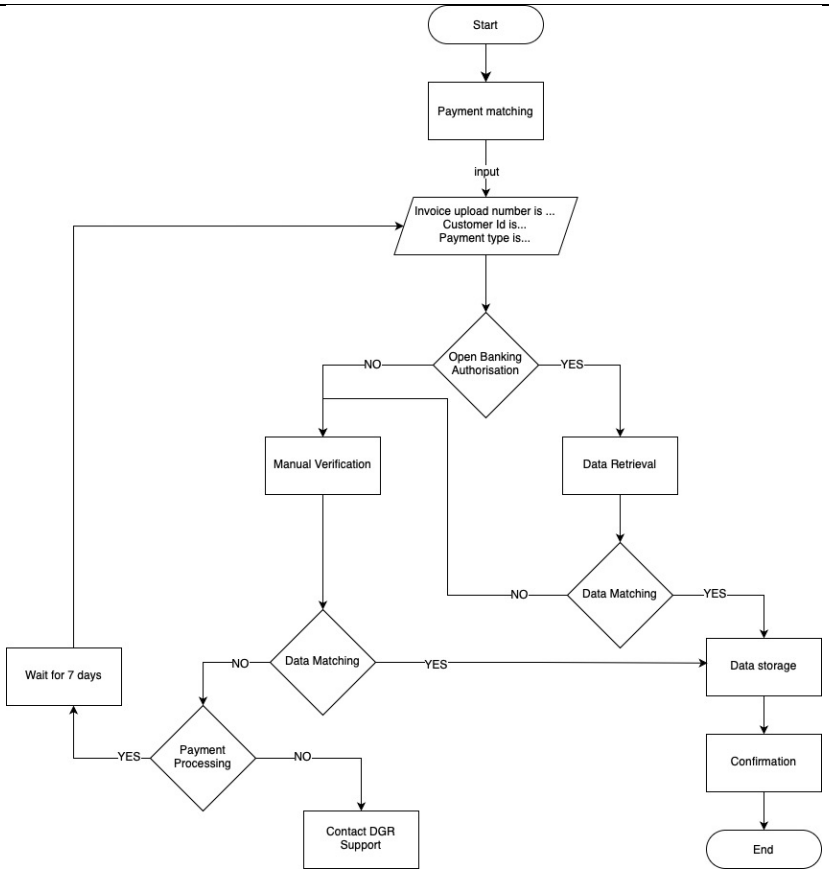
Use Case Number	7
Use Case Type	Base
Use Case Name	Manage Data
Actor Involved	Users, DGR platform, and System Administrator
Associated Data Source	DGR DB
Associated User Interfaces	Manage Data
Pre-Condition(s)	Customer must be a registered user with the DGR platform. And System must be operational.
Post Condition(s)	Customer's profile data can be updated in the system. This may include Personal details, Account details, Chosen charity, Contact preferences, and Payment details. All the data will be backed up and restored securely in the system, Plus, backups are scheduled and completed successfully. Upgrades are applied seamlessly, minimising downtime. Also, System can recover data in case of failure.
Detailed Description	Once received the data, new data (e.g., invoices, transactions, user accounts) generated during system operations is securely stored in the system's database. Automated backups are scheduled at regular intervals to create copies of the system data. These backups can be used to restore data in case of system failures or data corruption. The system periodically checks for available software upgrades from the development team. These upgrades might include bug fixes, security patches, or new functionalities. The administrator decides whether to install the upgrades based on their risk assessment and potential impact. The administrator performs regular tests of the data backup and recovery procedures to ensure they function correctly. This ensures the system can be restored to a previous state if a failure occurs.
Normal Course of Events Using a Flow Chart Diagram	

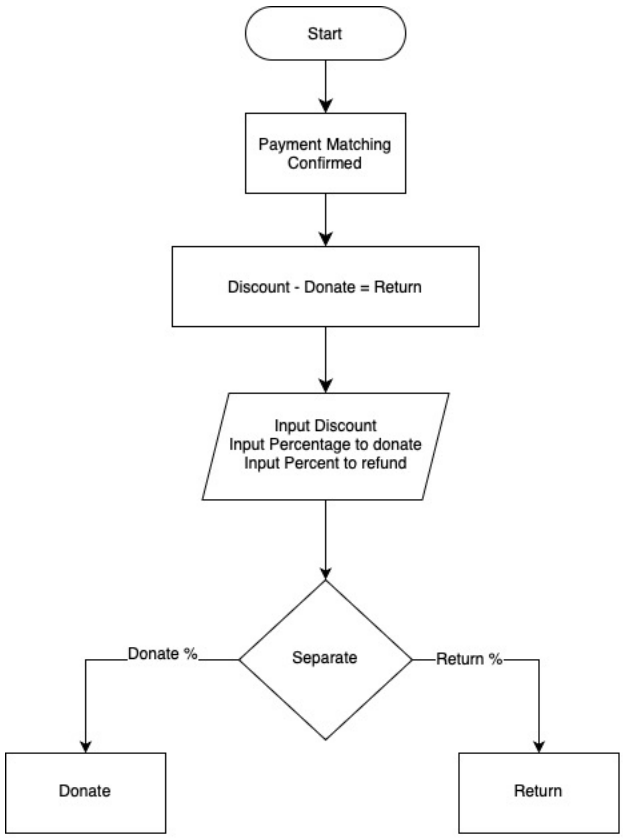
	
Alternate Course(s)	If a scheduled backup fails, the administrator receives an alert and must initiate a manual backup. If a software upgrade introduces unexpected issues, the administrator might need to roll back to the previous version. If a data recovery attempt fails, the administrator might need to consult technical support for assistance.

Use Case Number	8
Use Case Type	Base
Use Case Name	Monitor Systems Performance
Actor Involved	DGR platform, and System Administrator, and DGR support
Associated Data Source	DGR DB
Associated User Interfaces	Monitor Systems Performance
Pre-Condition(s)	DGR platform is operational and monitoring tools are active and configured
Post Condition(s)	System performance metrics are collected and analysed Track uptime. And, alerts are generated for any performance issues, delivered to DGR support.
Detailed Description	System continuously monitors various performance metrics (e.g., response times, resource utilisation, error rates). Performance data is analysed to identify potential bottlenecks or areas for improvement. If performance metrics exceed predefined thresholds, alerts are generated to notify technical staff via DGR support
Normal Course of Events Using a Flow Chart Diagram	
Alternate Course(s)	If monitoring tools malfunction, corrective actions are taken to restore functionality. If alerts indicate a security breach, appropriate security protocols are initiated

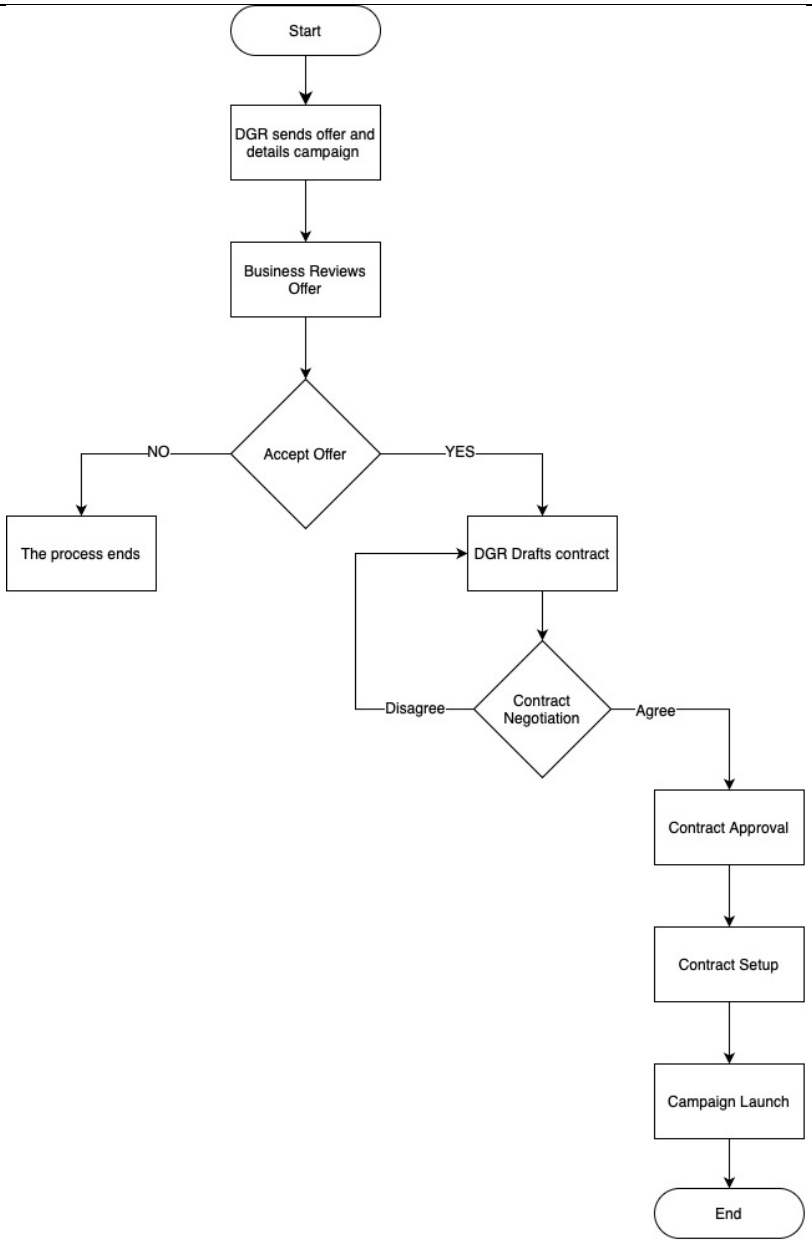
Use Case Number	9
Use Case Type	Base
Use Case Name	Manage Level Access in the systems
Actor Involved	Users, DGR platform, and System Administrator
Associated Data Source	DGR DB
Associated User Interfaces	Manage Level Access in the systems
Pre-Condition(s)	User must be logged in with an administrative account. And System access control mechanisms must be functional.
Post Condition(s)	User access levels are successfully modified within the system.
Detailed Description	System administrators access the user management function with the DGR platform this function allows them to view a list of existing users and their current access level, they can check by select a specific user and modify their access level, then confirm action and send notification.
Normal Course of Events Using a Flow Chart Diagram	
Alternate Course(s)	If the user attempting to manage access levels doesn't have the necessary administrative privileges, the system displays an error message denying access.

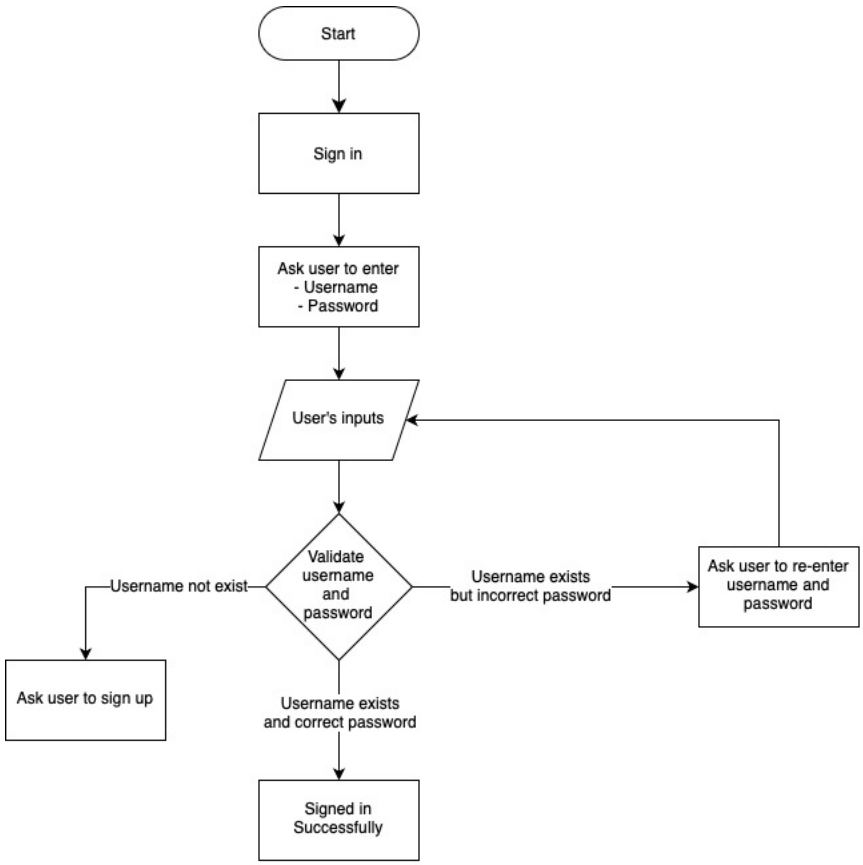
Use Case Number	10
Use Case Type	Base
Use Case Name	Handle Payment Matching
Actor Involved	Registered Customer, Merchant, Open Banking Provider, and DGR support
Associated Data Source	DGR DB
Associated User Interfaces	Handle Payment Matching
Pre-Condition(s)	Customer has made a full payment for an invoice through a participating merchant. The merchant has uploaded the corresponding invoice to the DGR platform. The customer's bank account information is accessible for verification through Open Banking with customer consent
Post Condition(s)	The DGR platform verifies the invoice details with the customer's payment. The system calculates the discount based on pre-defined rules or merchant-provided information. If a discount applies, a refund process will be used to transfer the discount amount back to the customer's payment method. The DGR platform records the transaction details (payment, invoice, discount, refund) for future reference.
Detailed Description	The DGR platform receives notification of a customer payment for an invoice uploaded by a participating merchant. If they consent to open banking, the system will verify the invoice details and calculate the applicable discount for the customer. If not, the administrator will verify manually. All the payment processing record need to wait 7 days and be rechecked again. For other issues, contact DGR support for help.
Normal Course of Events Using a Flow Chart Diagram	

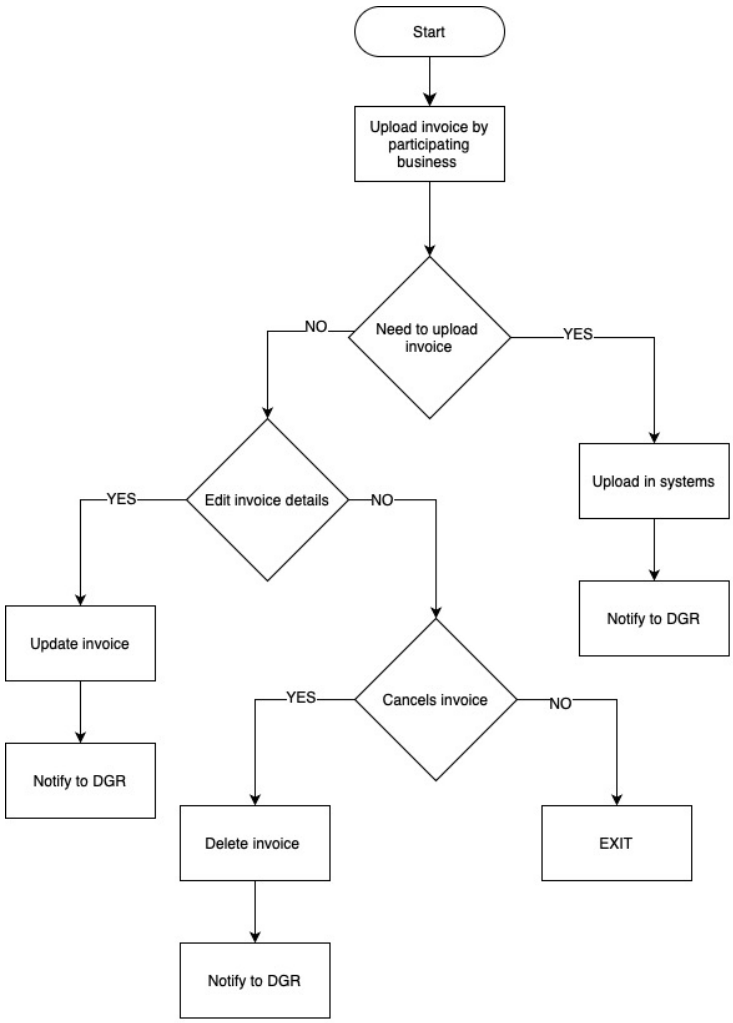
	
Alternate Course(s)	If the invoice verification fails (e.g., invoice amount mismatch), the system notifies the merchant and potentially holds the refund until the discrepancy is resolved

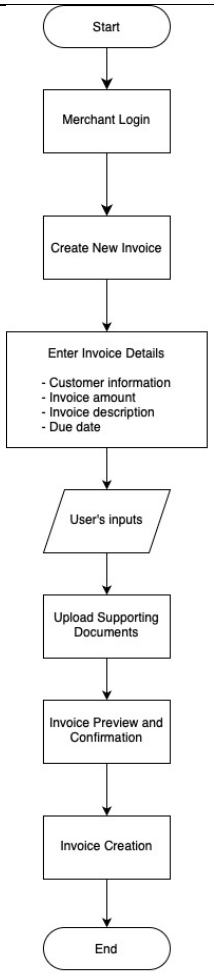
Use Case Number	11
Use Case Type	Base
Use Case Name	Calculate discount and donation
Actor Involved	DGR support, Registered Customer, Merchant
Associated Data Source	DGR DB
Associated User Interfaces	Calculate discount and donation
Pre-Condition(s)	Customer must be registered account and paid with registered payment method. Merchant has uploaded a valid invoice for the customer. Open Banking connection is established.
Post Condition(s)	Discount amount is calculated based on pre-defined rules or merchant settings. The invoice is marked as processed in the system. Customer receives a notification with the calculated discount and donation details. Merchant receives a notification with the processed invoice and discount information. The donation amount is transferred to the chosen charity.
Detailed Description	After payment matching is confirmed. The donation amount is calculated as a percentage of the discount. The processed invoice and discount information are stored in the system.
Normal Course of Events Using a Flow Chart Diagram	
Alternate Course(s)	If the discount calculation results in a zero value, no donation is calculated, and the customer is notified

Use Case Number	12
Use Case Type	Base
Use Case Name	Write Contract
Actor Involved	DGR and Merchant and DGR Platform
Associated Data Source	DGR DB
Associated User Interfaces	Write Contract
Pre-Condition(s)	Merchant must understand the concept of the DGR campaign and agree to provide involved data to DGR platform. System has verified customer and merchant eligibility for the DGR campaign
Post Condition(s)	DGR and Merchant must sign agree on the contract and comply with the conditions of the transaction, including the full invoice to DGR platform. The contract details are stored securely within the DGR platform.
Detailed Description	Once Merchant received the campaign details, they need to offer discount of their product and send it back to DGR, this process will rotate in contract negotiation process until both are agreed. Then, the merchant logo will be set up and launched campaign on DGR platform
Normal Course of Events Using a Flow Chart Diagram	

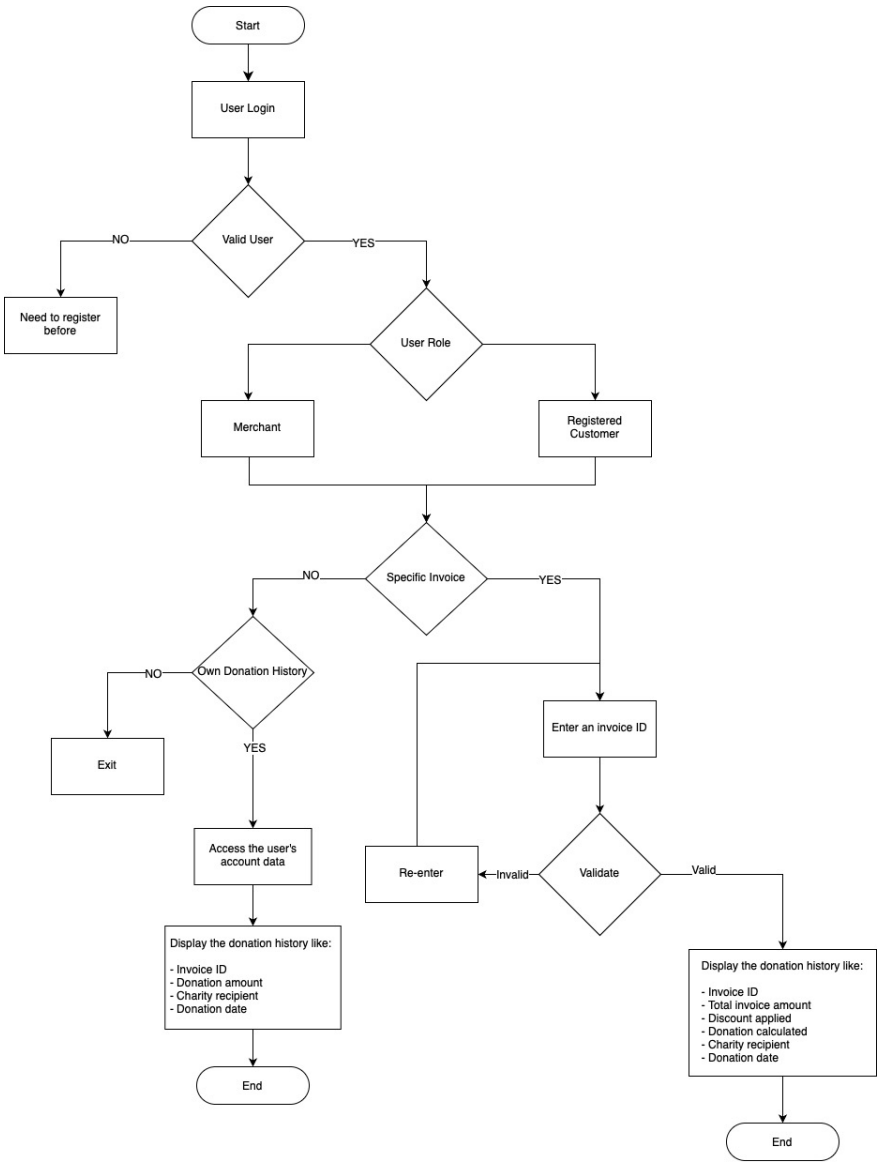
	
Alternate Course(s)	-

Use Case Number	13
Use Case Type	Include
Use Case Name	Sign in
Actor Involved	User, System Administrator
Associated Data Source	DGR DB
Associated User Interfaces	Sign in
Pre-Condition(s)	User account subsystems are available, and they have a registered account in the system.
Post Condition(s)	Once successfully authenticated, they can access the DGR platform, according to their level.
Detailed Description	Enters their login credentials (email address and password) on the DGR platform login page. The system validates the provided credentials against its user account database. If the credentials are valid, the system grants the customer access to their account dashboard and functionalities within the DGR platform. If not, the system will ask to sign up or re-enter.
Normal Course of Events Using a Flow Chart Diagram	
Alternate Course(s)	Security measures should be implemented to protect against unauthorised access attempts

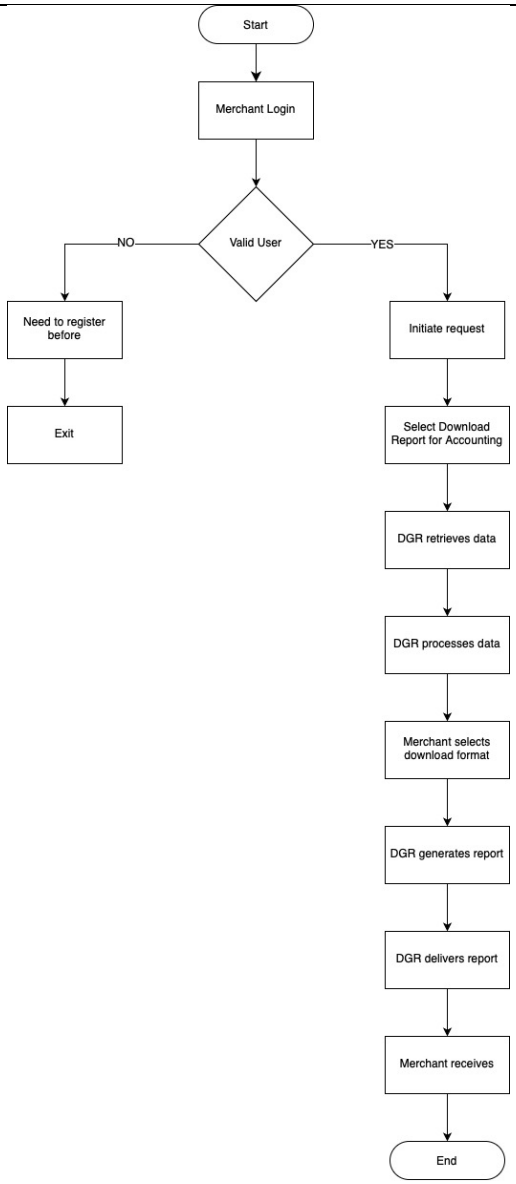
Use Case Number	14
Use Case Type	Base
Use Case Name	Upload Invoice
Actor Involved	Merchant, DGR Platform
Associated Data Source	DGR DB
Associated User Interfaces	Upload Invoice
Pre-Condition(s)	A participating merchant must be logged into the DGR platform. The product was paid for by registered customer.
Post Condition(s)	The uploaded invoice is stored securely in the DGR system along with relevant details (e.g., invoice amount, date, customer information). The system initiates the invoice verification process and step though calculate process to divide the discount after.
Detailed Description	The merchant initiates the "Upload Invoice" functionality and confirms it as a new invoice, edits invoice details, or cancels invoice.
Normal Course of Events Using a Flow Chart Diagram	
Alternate Course(s)	If the uploaded file format is invalid or exceeds the allowed size limit, the system prompts the merchant with an error message and instructions for uploading a valid file

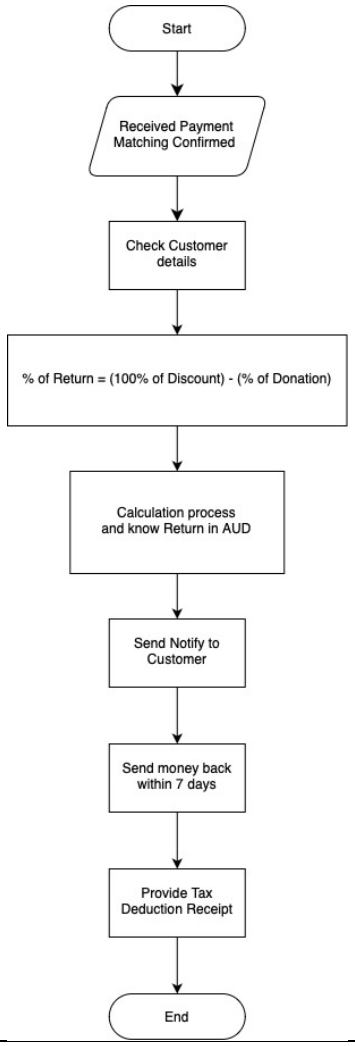
Use Case Number	15
Use Case Type	Include
Use Case Name	Create Invoice
Actor Involved	Merchant, Registered Customer, DGR Platform
Associated Data Source	DGR DB
Associated User Interfaces	Create Invoice
Pre-Condition(s)	A participating merchant must be logged into the DGR platform. The product was paid for by registered customer.
Post Condition(s)	A new invoice is created in the system, associated with the participating merchant. Invoice details are stored securely, including customer information, invoice amount, and relevant dates. The system may generate a unique invoice identifier for reference.
Detailed Description	The participating merchant initiates the "Create Invoice" process. The system prompts the merchant to enter invoice details, such as customer information, invoice amount, invoice due date, also supporting documents. Then preview the invoice and confirm invoice creation.
Normal Course of Events Using a Flow Chart Diagram	
Alternate Course(s)	If there are validation errors in the invoice data, the system displays error messages and prompts the merchant to correct the information before submitting again.

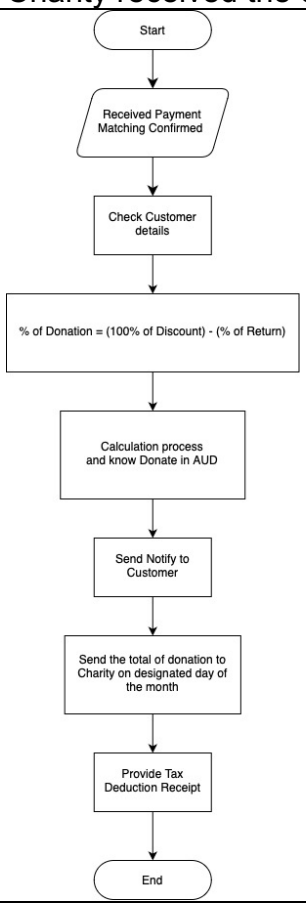
Use Case Number	16
Use Case Type	Base
Use Case Name	View Donation History
Actor Involved	Merchant, DGR support
Associated Data Source	DGR DB
Associated User Interfaces	View Donation History
Pre-Condition(s)	Merchant must have a registered account on the DGR platform. The DGR platform must be operational and accessible.
Post Condition(s)	The merchant can view a detailed breakdown of past donations facilitated through their DGR campaign, including, donation date, customer name, invoice amount, calculated discount, donation amount sent to charity
Detailed Description	The merchant accesses the DGR platform and navigates to the "Donation History" section. The system displays a list of all donations processed for the merchant's DGR campaign. Each donation entry provides details like date, customer information, invoice amount, discount applied, and the final donation amount sent to the chosen charity. Or they can check in specific invoice number.

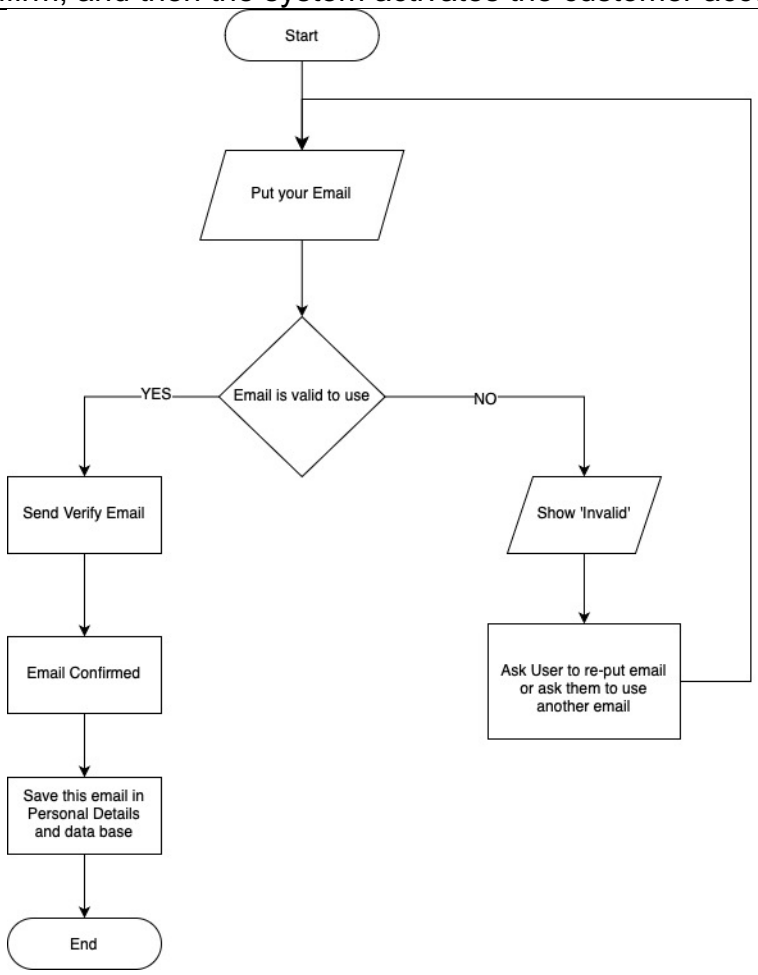
Normal Course of Events Using a Flow Chart Diagram	
Alternate Course(s)	If the merchant doesn't have any past donations, the system displays a message indicating there's no donation history yet. If there's a system error preventing access to donation history, the merchant receives an error message prompting them to contact DGR support.

Use Case Number	17
Use Case Type	Base
Use Case Name	Downloads Reports for accounting
Actor Involved	Merchant, DGR support
Associated Data Source	DGR DB
Associated User Interfaces	Downloads Reports for accounting
Pre-Condition(s)	Merchant must have a registered account on the DGR platform. The DGR platform must be operational and accessible. Relevant reports for the desired accounting period must be generated.
Post Condition(s)	The merchant successfully downloads a report in a suitable format (e.g., PDF, CSV) for accounting purposes. Plus, The downloaded report contains relevant information about DGR programme activity for the chosen period
Detailed Description	The merchant initiates a download request for a DGR programme report. The system authenticates the merchant and verifies their access level. The merchant specifies the desired accounting report type and timeframe (e.g., monthly, quarterly). The system retrieves the report data and allows the merchant to download it in a suitable format.

Normal Course of Events Using a Flow Chart Diagram	
Alternate Course(s)	If the desired report type or timeframe is unavailable, the system prompts the merchant to choose a valid option.

Use Case Number	18
Use Case Type	Include
Use Case Name	Refund Discount to Customer
Actor Involved	Registered Customer, Merchant, DGR, DGR platform
Associated Data Source	DGR DB
Associated User Interfaces	Refund Discount to Customer
Pre-Condition(s)	Customer has made a full payment for an invoice. Merchant has uploaded the invoice to the DGR platform. DGR platform has verified the invoice and confirmed they are matched by Open banking Provider.
Post Condition(s)	All discount applicable will be refunded to customer bank account as they were set in pay-out within 7 days. All the details will be stored in customer's profile and DGR system.
Detailed Description	Once, they confirmed the payment and invoice are matched, the system will calculate return discount in AUD. Notification Service will send an alert text to the customer about the payment will generate back within 7 days with tax receipt.
Normal Course of Events Using a Flow Chart Diagram	
Alternate Course(s)	If the refund process fails, the system notifies the DGR administrator for manual intervention.

Use Case Number	19
Use Case Type	Include
Use Case Name	Donate
Actor Involved	Registered Customer, Merchant, DGR, DGR platform, and Charity
Associated Data Source	DGR DB
Associated User Interfaces	Donate
Pre-Condition(s)	Customer has made a full payment for an invoice. Merchant has uploaded the invoice to the DGR platform. DGR platform has verified the invoice and confirmed they are matched by Open banking Provider.
Post Condition(s)	All the donate will be collected in one month and donate to Charity on a designated day. All the donate details are recorded in DGR platform database. Customer successfully donates a chosen amount to a designated charity.
Detailed Description	Once, they confirm the payment and invoice are matched, the system will calculate the donation in AUD. Notification Service will send an alert text to the customer and Merchant about the donation will send to Charity on which day and send email when the Charity received the donation.
Normal Course of Events Using a Flow Chart Diagram	
Alternate Course(s)	-

Use Case Number	20
Use Case Type	Extend
Use Case Name	Add Email Verification
Actor Involved	Customer, System administrator, DGR platform
Associated Data Source	DGR DB
Associated User Interfaces	Add Email Verification
Pre-Condition(s)	Customer has initiated account registration. Basic personal details, account details, chosen charity, and contact agreement have been provided.
Post Condition(s)	Customer's email address is verified. Customer account is created and activated when email verification succeeds.
Detailed Description	The customer chooses to add email verification during registration. The system will check if that email is valid to use or not, then send a verification email to the provided customer email address. The email contains a unique verification link and the customer needs to click the link to confirm, and then the system activates the customer account.
Normal Course of Events Using a Flow Chart Diagram	
Alternate Course(s)	If the customer clicks the verification link after a certain timeframe, the link expires, and the customer needs to request a new verification email.